Ex. No.	3	Name	Ajmal Rahman A
Date	18-10-2026	Reg. No.	192324094
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EXERCISE NAME

DEFECT COUNT MODELLING WITH THE POISSON DISTRIBUTION

SCENARIO

A manufacturing unit records the number of defective products found in different production batches. Model the defect counts using a suitable probability distribution and estimate the probability of observing a specified number of defects in a batch, so that the quality-control team can judge how unusual a given batch is.

ALGORITHM

Step 1 : Import the packages used in this exercise — NumPy for the random experiment, SciPy (stats) for the probability distributions and the chi-square test, and Matplotlib for the plots.

Step 2 : Record the number of defective products found in each of 200 inspected batches.

Step 3 : Examine which distribution suits the data. The recorded values are non-negative counts of a rare event observed in a fixed batch size, and the sample variance is nearly equal to the sample mean, which is the defining property of the Poisson distribution. The Poisson model is therefore adopted.

Step 4 : Estimate its parameter λ by the method of maximum likelihood, for which the estimator is the sample mean, $\hat{\lambda} = (1/n) \Sigma x_i$.

Step 5 : Build the frequency distribution of the observed counts for k = 0, 1, 2, …, 9 together with a last class of 10 or more, and compute for every class the fitted probability P(X = k) = e^(−λ) λ^k / k! and the expected number of batches, which is 200 × P(X = k).

Step 6 : Estimate the probability of observing the specified number of defects, P(X = 5), and the related probabilities P(X = 0), P(X ≤ 2), P(X ≥ 6), P(2 ≤ X ≤ 5) and P(X > 8), using the probability mass function and the cumulative distribution function of the fitted model.

Step 7 : Test how well the model fits with the chi-square goodness-of-fit statistic, $\chi^2 = \Sigma (O - E)^2 / E$, where the degrees of freedom are the number of classes reduced by one for the total and by one more for the estimated parameter λ. Compare the statistic with the critical value at the 5 % level of significance.

Step 8 : Compare the fitted Poisson probabilities with those of a binomial model Bin(n = 500, p = $\hat{\lambda}$ / 500), in which 500 items are produced in a batch and each item is defective with a small probability, to confirm that the Poisson model is the limiting form of the binomial for a large n and a small p.

Step 9 : Plot the observed relative frequency against the fitted Poisson probabilities as paired bars, and mark the specified number of defects on the plot.

Step 10 : Plot the empirical cumulative distribution of the recorded data against the theoretical Poisson cumulative distribution. Stop.

SOURCE CODE

Module 1 : Importing the Required Packages

```
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

plt.rcParams["font.family"]          = "serif"
plt.rcParams["font.serif"]           = ["Cambria", "Caladea", "DejaVu Serif"]
plt.rcParams["axes.spines.top"]      = False
plt.rcParams["axes.spines.right"]    = False
plt.rcParams["legend.frameon"]       = False
```

Module 2 : Recording the Defect Counts of the Production Batches

```
np.random.seed(42)
BATCHES   = 200                                  # number of batches inspected
LAMBDA_T  = 3.2                                 # true average defects per batch

defects = np.random.poisson(LAMBDA_T, BATCHES)   # defective items in each batch

print("Batches inspected :", BATCHES)
print("Defects in the first 20 batches :", defects[:20])
print("Total defective items :", defects.sum())
```

Module 3 : Choosing the Suitable Probability Distribution

```
mean_d = defects.mean()
var_d  = defects.var(ddof=1)
ratio  = var_d / mean_d

print("Sample mean     : %.4f" % mean_d)
print("Sample variance : %.4f" % var_d)
print("Variance / mean : %.4f  (close to 1 supports the Poisson model)" % ratio)

lam_hat = mean_d                          # maximum likelihood estimate of lambda
print("Estimated lambda (MLE) : %.4f" % lam_hat)
```

Module 4 : Frequency Distribution of the Observed Defect Counts

```
K_MAX  = 10
ks     = np.arange(0, K_MAX + 1)
obs    = np.array([np.sum(defects == k) for k in ks])
obs[-1] = np.sum(defects >= K_MAX)               # last class is "10 or more"

pmf    = stats.poisson.pmf(ks, lam_hat)
pmf[-1] = stats.poisson.sf(K_MAX - 1, lam_hat)   # tail probability P(X >= 10)
exp_f  = pmf * BATCHES

for k, o, p, e in zip(ks, obs, pmf, exp_f):
    print("k %2d | observed %3d | P(X=k) %.4f | expected %6.2f" % (k, o, p, e))
```

Module 5 : Probability of Observing a Specified Number of Defects

```
K_SPEC = 5                                      # the specified number of defects

queries = [
    ("Exactly %d defects (the specified number)" % K_SPEC, "P(X = %d)" % K_SPEC,
     stats.poisson.pmf(K_SPEC, lam_hat)),
    ("A defect-free batch",      "P(X = 0)",
     stats.poisson.pmf(0, lam_hat)),
```

```
    ("At most 2 defects",        "P(X ≤ 2)",
     stats.poisson.cdf(2, lam_hat)),
    ("At least 6 defects",       "P(X ≥ 6)",
     stats.poisson.sf(5, lam_hat)),
    ("Between 2 and 5 defects",  "P(2 ≤ X ≤ 5)",
     stats.poisson.cdf(5, lam_hat) - stats.poisson.cdf(1, lam_hat)),
    ("More than 8 defects",      "P(X > 8)",
     stats.poisson.sf(8, lam_hat)),
]
for desc, expr, p in queries:
    print("%-42s %-16s = %.6f  (%.3f %%)" % (desc, expr, p, 100*p))

rel_freq = np.mean(defects == K_SPEC)
print("Relative frequency of exactly %d defects : %.4f" % (K_SPEC, rel_freq))
```

Module 6 : Goodness-of-Fit Test for the Poisson Model

```
chi_sq = float(np.sum((obs - exp_f) ** 2 / exp_f))
dof    = len(ks) - 1 - 1                    # classes - 1 - estimated parameters
p_val  = stats.chi2.sf(chi_sq, dof)
crit   = stats.chi2.ppf(0.95, dof)

print("Chi-square statistic : %.4f" % chi_sq)
print("Degrees of freedom   : %d" % dof)
print("Critical value (5 %%) : %.4f" % crit)
print("p-value              : %.4f" % p_val)
print("Decision :", "Poisson accepted" if chi_sq < crit else "Poisson rejected")
```

Module 7 : Comparison with the Binomial Model

```
n_items = 500                                    # items produced in each batch
p_item  = lam_hat / n_items              # probability that an item is defective
bin_pmf = stats.binom.pmf(ks, n_items, p_item)

print("Binomial(n = %d, p = %.6f) mean : %.4f"
      % (n_items, p_item, n_items*p_item))
print("Maximum |Poisson - Binomial| over k = 0..10 : %.6f"
      % np.max(np.abs(pmf[:-1] - bin_pmf[:-1])))
```

Module 8 : Visualisation of the Fitted Distribution

```
fig, ax = plt.subplots(figsize=(6.6, 3.4))
ax.bar(ks - 0.18, obs / BATCHES, width=0.36, color="#2F6DB5",
       label="Observed relative frequency")
ax.bar(ks + 0.18, pmf, width=0.36, color="#C0702A",
       label="Poisson P(X = k),  $\\lambda$ = %.3f" % lam_hat)
ax.axvline(K_SPEC, color="#4E8F59", lw=1.3, ls="--")
ax.text(K_SPEC + 0.15, max(obs / BATCHES) * 0.92,
        "specified k = %d" % K_SPEC, fontsize=8, color="#4E8F59")
ax.set_xlabel("Number of defective products in a batch (k)")
ax.set_ylabel("Probability")
ax.set_xticks(ks, [str(k) for k in ks[:-1]] + ["10+"])
ax.grid(axis="y", alpha=0.55);  ax.set_axisbelow(True)
ax.legend(loc="upper right", fontsize=8)
fig.tight_layout()
fig.savefig("plot1_pmf.png", dpi=300, bbox_inches="tight")

fig, ax = plt.subplots(figsize=(6.6, 3.4))
emp_cdf = np.cumsum(obs) / BATCHES
ax.step(ks, emp_cdf, where="mid", color="#2F6DB5", lw=1.6,
        label="Empirical CDF of the 200 batches")
ax.step(ks, np.cumsum(pmf), where="mid", color="#C0702A", lw=1.6, ls="--",
        label="Poisson CDF,  $\\lambda$ = %.3f" % lam_hat)
ax.scatter(ks, emp_cdf, s=14, c="#2F6DB5")
ax.set_xlabel("Number of defective products in a batch (k)")
ax.set_ylabel("Cumulative probability  P(X $\\leq$ k)")
```

```
ax.set_xticks(ks);  ax.set_ylim(0, 1.05)
ax.set_yticks(np.arange(0, 1.01, 0.2))
ax.grid(alpha=0.55);  ax.set_axisbelow(True)
ax.legend(loc="lower right", fontsize=8)
fig.tight_layout()
fig.savefig("plot2_cdf.png", dpi=300, bbox_inches="tight")
```

OUTPUT

Table 1 - Recorded Defect Counts of the First 15 Batches

S.No	Batch ID	Defective Products	Batch ID	Defective Products
1	B001	4	B016	1
2	B002	1	B017	3
3	B003	3	B018	2
4	B004	3	B019	4
5	B005	2	B020	0
6	B006	3	B021	4
7	B007	3	B022	2
8	B008	3	B023	2
9	B009	2	B024	5
10	B010	4	B025	3
11	B011	2	B026	4
12	B012	1	B027	1
13	B013	4	B028	4
14	B014	6	B029	3
15	B015	2	B030	3

Table 2 - Frequency Distribution and the Fitted Poisson Probabilities

Defects k	Observed Batches	Observed Rel. Frequency	Poisson P(X = k)	Expected Batches
0	10	0.0500	0.0435	8.70
1	29	0.1450	0.1364	27.27
2	37	0.1850	0.2138	42.75
3	50	0.2500	0.2234	44.68
4	28	0.1400	0.1751	35.02
5	23	0.1150	0.1098	21.95
6	17	0.0850	0.0574	11.47
7	4	0.0200	0.0257	5.14
8	1	0.0050	0.0101	2.01
9	1	0.0050	0.0035	0.70
10 or more	0	0.0000	0.0015	0.30
Total	200	1.0000	1.0000	200.00

Table 3 - Parameter Estimation and Justification of the Poisson Model

Parameter	Value	Remark
Number of batches (n)	200	Sample size
Total defective products	627	Sum of all batches
Minimum defects in a batch	0	Non-negative count
Maximum defects in a batch	9	Non-negative count
Sample mean	3.1350	Estimator of λ
Sample variance	3.2028	Estimator of λ
Variance / mean ratio	1.0216	Close to 1 → Poisson is suitable
Estimated λ (MLE)	3.1350	$\hat{\lambda}$ = sample mean
Distribution adopted	Poisson(λ = 3.135)	Counts of rare defects per batch

Table 4 - Required Probabilities from the Fitted Poisson Model

S.No	Probability Required	Expression	Value	Percentage	Expected Batches out of 200
1	Exactly 5 defects (the specified number)	$P(X = 5)$	0.109772	10.977 %	22.0
2	A defect-free batch	$P(X = 0)$	0.043500	4.350 %	8.7
3	At most 2 defects	$P(X \leq 2)$	0.393634	39.363 %	78.7
4	At least 6 defects	$P(X \geq 6)$	0.098136	9.814 %	19.6
5	Between 2 and 5 defects	$P(2 \leq X \leq 5)$	0.721993	72.199 %	144.4
6	More than 8 defects	$P(X > 8)$	0.005026	0.503 %	1.0

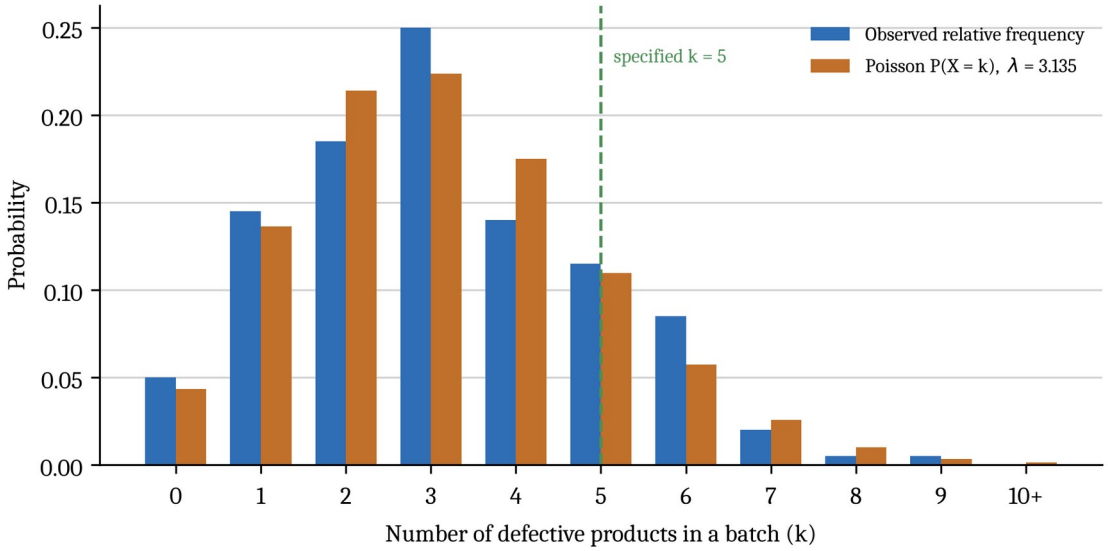
Table 5 - Chi-Square Goodness-of-Fit Test for the Poisson Model

Parameter	Value
Null hypothesis H_0	The defect counts follow a Poisson distribution
Number of classes	11
Estimated parameters	1 (λ)
Degrees of freedom	9
Chi-square statistic χ^2	7.0248
Critical value at 5 % level	16.9190
p-value	0.6345
Decision	χ^2 < critical value → H_0 accepted
Conclusion	The Poisson model fits the recorded defect data

Table 6 - Comparison of the Poisson and Binomial Models

Defects k	Poisson P(X = k)	Binomial P(X = k)	Difference
0	0.043500	0.043073	+0.000427
1	0.136372	0.135884	+0.000487
2	0.213763	0.213914	-0.000151
3	0.223382	0.224051	-0.000669
4	0.175076	0.175648	-0.000572
5	0.109772	0.109939	-0.000167
6	0.057356	0.057228	+0.000128
7	0.025687	0.025482	+0.000205
8	0.010066	0.009908	+0.000158
9	0.003506	0.003418	+0.000089

Plot 1 - Observed Relative Frequency against the Fitted Poisson Probabilities



Plot 2 - Empirical and Theoretical Cumulative Distribution Functions

